

Basics

TS components

- trend
- seasonality
- cyclicality
- random variation

ACF/PACF

- signif autocorrelation?
- can you detect trend/seasonality?
- what is ACF/PACF?
- positive vs negative ACF?

stationarity

- constant mean, variance?
- transformation needed?
- White Noise $\Rightarrow$ stationary, $N(0, \sigma^2)$
- KPSS test

NB for ARIMA, not so much for the rest

MA smoothing

- CMA or MA required?
- length of seasonal variation?
- # of seasons ($= k$ or g)
- calculate! for transform.
- pros and cons?

Transformations

- SD: log, BoxCox, power
- Mean: seasonality? trend?
- lag g lag 1
- first-order / second-order?

Residuals

- plot: pattern?
- ACF plot: signif autocor?
- Histogram/QQ-plot: normal?
- Ljung-Box test
 - H_0, H_1 ? Interpret?
 - Test stat from scratch? $Q_{Lg}(m, k)$

we want it to look like a white noise process

Modelling

* TS Model: multiplicative: $Y_t = T_t \times S_t \times R_t, C_t = 1$

1 Classical decomp

decompose to better understand underlying comp.

practice obtaining table of values:

- MA or CMA
- dettrend: Y_t / MA
- find AGI (Correction factor)
- discuss / interpret AGIs
- de-seasonalize: Y_t / AGI

- $R_t = \frac{Y_t}{\hat{S}_t \times \hat{CMA}(k)_t}$
- linear trend? regression?

$\Rightarrow Y_t = (\hat{\beta}_0 + \hat{\beta}_1 t) \hat{S}_t$
used for forecasting?

2 Simple forecasting

average / naïve

drift

seasonal naïve

- best long-term / short-term?
- Why?

Forecasting accuracy

- test vs train data?
- MAE + MSE / RMSE
- only interested in TEST values: smaller = better
- prediction intervals?
- AIC model fit?

2 Exp smoothing

SES

- no trend, no seasonality

Holt-Winters

- trend and seasonality
- component form?
- significance of large vs. small smoothing parameters?

PRACTICE obtaining table values with the component form equations!

- How are estimates found, minimize SSE?

3 ARIMA

- $AR(p) + I(d) + MA(q)$
- Describe each component?
- How is MA different from AR?
- Steps for modelling?
- size, sign of AR, MA parameters - significant?
- AR(1) mean, variance?
- MA(1) mean, variance?
- AR(p): look at PACF
- MA(q): look at ACF
- Why do we fit a range of low-order models?
- which model is best?
- calculate AIC?
- calculate constant model term?
- investigate residuals of fitted models?